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DEPARTMENT OF PHYSICS

RV COLLEGE OF ENGINEERING®

(AUTONOMOUS INSTITUTION AFFILIATED TO VTU, BELAGAVI)

ENGINEERING PHYSICS

PRACTICAL FILE

(ME & CV STREAM)

COURSE CODE: PHY211BI / PHY211DI

2023-24

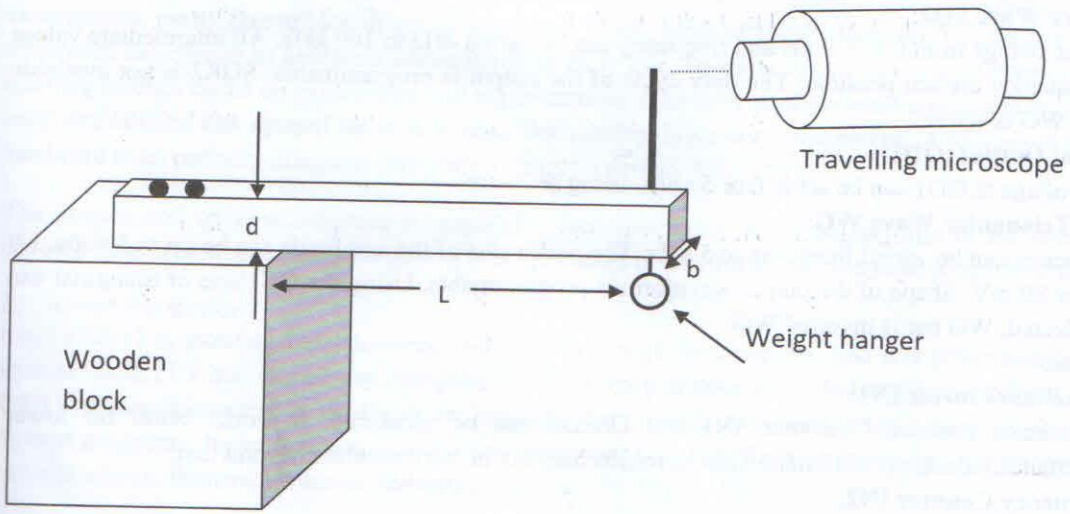
For the First Semester B.E

Name of the student	
Section, Batch	
Program	
Roll No. / USN	
Faculty In-charge	**1.**
	2.

SINGLE CANTILEVER

OBSERVATIONS:

Experimental Setup:



Formula:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Where, m = mass for which depression produced is 0.04 kg (40 gm),

L is the length of the cantilever in metre

b is the breadth of the cantilever in metre

d is the thickness of the cantilevering metre

δ is the mean depression for 40g in metre

g is the acceleration due to gravity = 9.8 m/s

Least count of the travelling microscope

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.05}{50} = 0.001 \text{ c m}$$

SINGLE CANTILEVER

Experiment No: Date:

Aim: To determine the Young's modulus of the material of the given metal strip.

Apparatus: Thick rectangular metallic strip (cantilever), slotted weights with hanger, travelling microscope, screw gauge, Vernier callipers and metre scale etc.

Principle: Young's modulus, which is one of the elastic constants, is defined as the ratio of longitudinal stress to the longitudinal strain within elastic limit. For a given strip, the depression produced at the loaded end of the cantilever depends on the load and on the distance from the fixed end. This is measured to calculate the Young's modulus (q) of the material.

Formula:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Where, m = mass for which depression produced is 0.04 kg (40 gm),

L is the length of the cantilever in metre

b is the breadth of the cantilever in metre

d is the thickness of the cantilevering metre

δ is the mean depression for 40g in metre

g is the acceleration due to gravity = 9.8 m/s

Procedure:

- Suspend the weight hanger with mass W, at the bottom of the free end of the metallic strip (into a metal loop fixed below the pin).
- Adjust the vertical traverse of travelling microscope to focus the tip of the pin with the horizontal cross wire or the point of the intersection of cross wires.
- Note down the reading of the travelling microscope on the vertical scale when the load is W in table 1.
- Now add a mass of 10 g to the weight hanger and adjust the travelling microscope using the fine motion screw (vertical motion) to focus the tip of the pin with the point of intersection of cross wires. Note down the travelling microscope reading (for a load of W+10).

Paste the data sheets here

Table 1: Travelling microscope readings for the load increasing.

Load (g)	MSR(cm)	CVD	TR= MSR + (CVD x LC) (cm)
W	6.10	18	6.10 + (18×0.001) = 6.118
W + 10	6.00	19	6.00 + (19×0.001) = 6.019
W + 20	5.90	27	5.90 + (27×0.001) = 5.927
W + 30	5.80	41	5.80 + (41×0.001) = 5.841
W + 40	5.70	46	5.70 + (46×0.001) = 5.746
W + 50	5.65	13	5.65 + (13×0.001) = 5.663
W + 60	5.55	23	5.55 + (23×0.001) = 5.573
W + 70	5.45	39	5.45 + (39×0.001) = 5.489

Table 2: Travelling microscope readings for the load decreasing.

Load (g)	MSR(cm)	CVD	TR= MSR + (CVD x LC) (cm)
W + 70	5.45	37	5.45 + (37×0.001) = 5.487
W + 60	5.55	25	5.55 + (25×0.001) = 5.575
W + 50	5.65	14	5.65 + (14×0.001) = 5.664
W + 40	5.70	43	5.70 + (43×0.001) = 5.743
W + 30	5.80	38	5.80 + (38×0.001) = 5.838
W + 20	5.90	24	5.90 + (24×0.001) = 5.924
W + 10	6.00	18	6.00 + (18×0.001) = 6.018
W	6.10	14	6.10 + (14×0.001) = 6.114

Table 3: Mean depression and depression for a load of 40g (difference column)

Load (g)	TM reading (cm)		Mean TMR $R_L = (R_1 + R_2)/2$	Load (g)	TM readings (cm)		Mean TMR $R_R = (R_3 + R_4)/2$	Depression δ for 40g ($\delta = R_L \sim R_R$) cm
	Load increasing R_1	Load Decreasing R_2			Load Increasing R_3	Load decreasing R_4		
W	6.118	6.114	6.116	W + 40	5.746	5.743	5.7445	0.3545
W + 10	6.019	6.018	6.0185	W + 50	5.663	5.664	5.6635	0.3550
W + 20	5.927	5.924	5.9255	W + 60	5.573	5.575	5.574	0.3515
W + 30	5.841	5.838	5.8395	W + 70	5.489	5.487	5.488	0.3515

Mean depression for a load of 40 g = δ = 0.3531 cm.

- Repeat the procedure by increasing the load in the weight hanger in steps of 10g up to a maximum load of W+70.

- Repeat the same by decreasing the load in steps of 10g and note down the readings for W+70, W+60 up to W and enter the readings in table 2.

- Tabulate the readings of increasing and decreasing loads in table 3. Compute the mean value of the reading corresponding to each load and find the depression 'δ' for a load of 40 g (m) by a difference column method.

- Measure the length 'L' of the cantilever from the edge of the wooden block to the position of the pin using a metre scale.

- Determine the breadth 'b' of the cantilever using vernier callipers at four different places on the metal strip and calculate the mean breadth.

- Determine the thickness 'd' of the cantilever using screw gauge at four different places on the metal strip and find the mean thickness.

- Compute the Young's modulus of the material of the cantilever by substituting the values of m, L, b, d and δ in the given formula.

Note:

1. Level the travelling microscope using a spirit level.
2. Once you start the experiment do not shake or lean on the table, as this will disturb the focusing and affect the reading.
3. Add/ remove the weights gently on to/from the weight hanger
4. While performing the experiment care is to be taken to rotate the fine motion screw in only one direction so as to avoid backlash error.

Paste the data sheets here

Length of the cantilever, L = 21.7 cm.

Table 4: Breadth of the cantilever using vernier callipers

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.1}{10} = 0.01 \text{ cm}$$

Trial No.	MSR(cm)	CVD	TR = MSR + (CVD x LC) (cm)
1	2.6	9	2.6 + (9 x 0.01) = 2.69
2	2.6	8	2.6 + (8 x 0.01) = 2.68
3	2.7	0	2.7 + (0 x 0.01) = 2.70
4	2.6	9	2.6 + (9 x 0.01) = 2.69

Mean Breadth b = 2.69 cm

Table 5: Thickness (d) of the cantilever using screw gauge.

Least Count of Screw Gauge:

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{1}{1} = 1 \text{ mm}$$

$$L.C = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) =

Trial No.	PSR(mm)	HSD	TR= PSR + (HSD- ZE)LC (mm)
1	0	93	0 + [93 - (-3)] x 0.01 = 0.96
2	0	95	0 + [95 - (-3)] x 0.01 = 0.98
3	0	96	0 + [96 - (-3)] x 0.01 = 0.99
4	0	95	0 + [95 - (-3)] x 0.01 = 0.98

Mean thickness, d = 0.9775 mm

Result: Young's modulus of the given material is found to be 1.8060×10^{11} N m^{-2}

CALCULATIONS:

Mean TMR

$W = \frac{6.118 + 6.114}{2} = 6.116$ cm

$W+10 = \frac{6.019 + 6.018}{2} = 6.0185$ cm

$W+20 = \frac{5.927 + 5.924}{2} = 5.9255$ cm

$W+30 = \frac{5.841 + 5.838}{2} = 5.8395$ cm

$W+40 = \frac{5.746 + 5.743}{2} = 5.7445$ cm

$W+50 = \frac{5.663 + 5.664}{2} = 5.6635$ cm

$W+60 = \frac{5.573 + 5.575}{2} = 5.574$ cm

$W+70 = \frac{5.489 + 5.487}{2} = 5.488$ cm

Depression δ

$\delta_1 = 6.116 - 5.7445 = 0.3545$ cm

$\delta_2 = 6.0185 - 5.6635 = 0.3550$ cm

$\delta_3 = 5.9255 - 5.574 = 0.3515$ cm

$\delta_4 = 5.8395 - 5.488 = 0.3515$ cm

Mean Depression, $\delta = \frac{0.3545 + 0.3550 + 0.3515 + 0.3515}{4} = \frac{1.4125}{4} = 0.3531$ cm

Mean breadth, $b = \frac{2.69 + 2.68 + 2.70 + 2.69}{4} = \frac{10.76}{4} = 2.69$ cm

Mean thickness, $d = \frac{0.96 + 0.98 + 0.99 + 0.98}{4} = \frac{3.91}{4} = 0.9775$ cm

$$Y = \frac{4mgL^3}{bd^3\delta_{mean}} = \frac{4 \times 0.04 \times 9.8 \times (21.7 \times 10^{-2})}{(2.69 \times 10^{-2}) \times (0.9775 \times 10^{-3})^3 \times (0.3531 \times 10^{-2})}$$

$$= \frac{1.568 \times 10218.313 \times 10^{-6}}{0.949839 \times 10^{-4} \times (0.9775 \times 10^{-3})^3}$$

$$= 18060.30 \times 10^{-6} \times 10^{13}$$

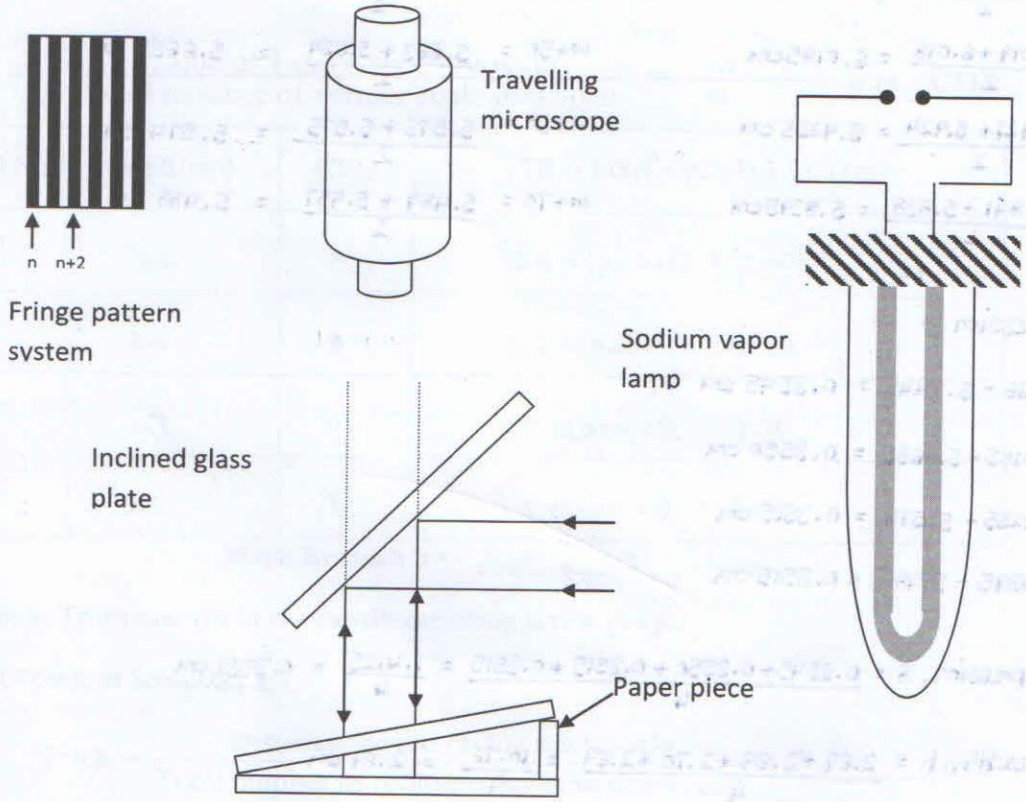
$$= 1.8060 \times 10^{11} \text{ N/m}^2$$

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Result: Young's modulus of the given material is found to be 1.8060×10^{11} N m^{-2}

INTERFERENCE AT A WEDGE

OBSERVATIONS



Least count of the travelling microscope

$$LC = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.05}{50} \text{ cm} = 0.001 \text{ cm}$$

Tabular column 1

Fringe number	TM readings, MSR + (CVD x LC)	Total reading (cm)
N	3.15 + (18 x 0.001)	3.168
n + 2	3.20 + (8 x 0.001)	3.208
n + 4	3.25 + (4 x 0.001)	3.254
n + 6	3.30 + (0 x 0.001)	3.300
n + 8	3.30 + (36 x 0.001)	3.336
n + 10	3.35 + (23 x 0.001)	3.373
n + 12	3.40 + (12 x 0.001)	3.412
n + 14	3.45 + (2 x 0.001)	3.452
n + 16	3.50 + (4 x 0.001)	3.504
n + 18	3.50 + (31 x 0.001)	3.531
n + 20	3.55 + (19 x 0.001)	3.569
n + 22	3.60 + (6 x 0.001)	3.606

INTERFERENCE AT A WEDGE

Experiment No. Date:

Aim: To determine the thickness of a thin paper piece by setting up interference fringes at an air wedge.

Apparatus: Two (optically flat) thick glass plates, sodium vapour lamp, travelling microscope (bridge type), a thin paper piece, reflecting glass plate, reading lens and a scale.

Principle: When a piece of paper is placed between two optically plane glass plates, a wedge shaped air film is formed between the two plates. If a parallel beam of monochromatic light is made to incident normally on the wedge (upper glass plate), the beam gets refracted through the upper glass plate. Two reflected rays from the upper and lower positions of the air film form two coherent sources. As the two portions in the reflected beam, are derived from the same incident beam and have path difference, they produce interference. The interference pattern, when viewed through a travelling microscope, appears as equally spaced alternate dark and bright straight fringes, parallel to the edge of the wedge.

Procedure:

a) Experimental Setup

The given optically plane glass plates and the reflecting glass plate are cleaned well by using muslin cloth before starting the experiment. The optically plane glass plates are kept one above the other. A thin rectangular paper piece is inserted between the glass plates at one extreme end, such that the piece of paper is perpendicular to the length of the glass plate. Now a wedge shaped air film is formed in between the plates. The thickness of the air film gradually increases, from the point of contact of glass plate, till the inner edge of the paper piece.

Now illuminate the air wedge by a parallel beam of sodium light using reflecting glass plate by making an angle of inclination 45° to the horizontal as shown in the diagram. The whole setup is kept below the objective of the travelling microscope such that the illuminated portion of the air wedge comes under the field of view of travelling microscope. Then focus the travelling microscope using rack and pinion screw such that the alternate dark and bright interference fringes are clearly seen.

b) Measurement of fringe width

The least count of travelling microscope is determined by finding the ratio of the value of one main scale division to the total number of vernier scale divisions.



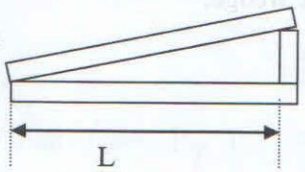
Tabular column 2

Fringe No.	TM Reading (cm)	Fringe No.	TM Reading (cm)	Width of 12 fringes (cm)
N	3.168	n + 12	3.412	0.244
n + 2	3.208	n + 14	3.452	0.244
n + 4	3.254	n + 16	3.504	0.250
n + 6	3.300	n + 18	3.531	0.231
n + 8	3.336	n + 20	3.569	0.233
n + 10	3.373	n + 22	3.606	0.233

Average width of 12 fringes, x = 2.325×10^{-3} m

The fringe width, β, is x/12 = 1.93×10^{-4} m.

To measure the length, L, of the air wedge



L = 0.04 m

The thickness of the paper is calculated by using the formula

$$t = \frac{\lambda L}{2\beta} = 6.1067 \times 10^{-5} \text{ m}$$

Here
t is the thickness of the paper ---------- m
L is the length of the air wedge ---------- m
λ is the wavelength of sodium light = 589.3nm
β is the fringe width ---------m

The vertical cross wire is adjusted to be at the center of any one of the dark fringes. The reading on the main scale and the coinciding vernier scale division are noted. Let this fringe be the nth dark fringe.

Next the reading corresponding to the dark fringes n+2, n+4, n+6 up to n+22 are noted. The microscope may be moved either to the left or the right side for this purpose.

The readings are tabulated in the observation column and the width of 12 dark fringes is calculated. The average width of 12 dark fringes and hence the fringe width, β, is calculated.

The length of the air wedge between the point of contact of the glass plates and the inner edge of the paper piece is measured using traveling microscope / scale.

The thickness of the given paper is calculated using the formula.

$$t = \frac{\lambda L}{2\beta} \text{ m}$$

$$t = \frac{589.3 \times 10^{-9} \times 4 \times 10^{-2}}{2 \times 1.93 \times 10^{-4}}$$

$$t = 6.1067 \times 10^{-5} \text{ m}$$

show more detailed steps here!

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Result:

The thickness of the given paper = 6.1067×10^{-5} m. → high value.

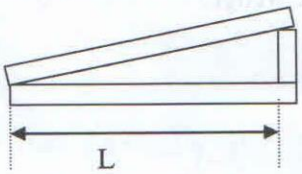
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The fringe width, β, is x/12 = 1.93×10^{-4} m.

To measure the length, L, of the air wedge



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The thickness of the paper is calculated by using the formula

$$t = \frac{\lambda L}{2\beta} = 6.1067 \times 10^{-5} \text{ m}$$

Here
t is the thickness of the paper ---------- m
L is the length of the air wedge ---------- m
λ is the wavelength of sodium light = 589.3nm
β is the fringe width ---------m

The vertical cross wire is adjusted to be at the center of any one of the dark fringes. The reading on the main scale and the coinciding vernier scale division are noted. Let this fringe be the nth dark fringe.

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The length of the air wedge between the point of contact of the glass plates and the inner edge of the paper piece is measured using traveling microscope / scale.

The thickness of the given paper is calculated using the formula.

$$t = \frac{\lambda L}{2\beta} \text{ m}$$

$$t = \frac{589.3 \times 10^{-9} \times 4 \times 10^{-2}}{2 \times 1.93 \times 10^{-4}}$$

$$t = 6.1067 \times 10^{-5} \text{ m}$$

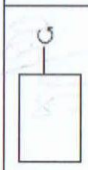
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Result:

The thickness of the given paper = 6.1067×10^{-5} m. → high value.

TORSION PENDULUM – OBSERVATIONS

Body	M (Kg)	Dimension (m)	Axis	Moment of Inertia I (Kgm²)	Time for 10 osc. (s)	Mean time 't' for 10 oscillation. (s)	Period T=t/10 (s)	$\frac{I}{T^2}$ (Kgm²s⁻²)
Circular plate	0.88kg	Radius = 0.076m		$I_1 = \frac{MR^2}{2}$ $= 2.54\times10^{-3}$	1. 35.63 2. 35.47 3. 35.94	35.68	$T_1 = 3.568$	$\frac{I_1}{T_1^2} = 1.99\times10^{-4}$
				$I_2 = \frac{MR^2}{4}$ $= 1.27\times10^{-3}$	1. 25.22 2. 25.44 3. 25.50	25.38	$T_2 = 2.538$	$\frac{I_2}{T_2^2} = 1.97\times10^{-4}$
Rectangular plate	1.015kg	Length, L = 15cm = 0.15m		$I_3 = \frac{M(L^2+B^2)}{12}$ $= 2.74\times10^{-3}$	1. 37.63 2. 37.84 3. 37.44	37.63	$T_3 = 3.763$	$\frac{I_3}{T_3^2} = 1.94\times10^{-4}$
				$I_4 = \frac{ML^2}{12}$ $= 1.90\times10^{-3}$	1. 31.78 2. 31.53 3. 31.47	31.59	$T_4 = 3.159$	$\frac{I_4}{T_4^2} = 1.97\times10^{-4}$
		Breadth, B = 10cm = 0.1m		$I_5 = \frac{MB^2}{12}$ $= 8.45\times10^{-4}$	1. 20.66 2. 20.62 3. 20.69	20.65	$T_5 = 2.065$	$\frac{I_5}{T_5^2} = 1.98\times10^{-4}$

$\left(\frac{I}{T^2}\right)_{\text{mean}} = 1.97\times10^{-4}$ Kgm²s⁻²

TORSION PENDULUM

Experiment No: Date:

Aim: To determine the moment of inertia of the given irregular body and Rigidity modulus of the material of the give wire.

Apparatus and other materials required: Rectangular, circular and irregularly shaped plates, steel or brass wire, chuck nuts, stop clock, pointer, metre scale, Screw gauge weight box.

Principle:

Torsion pendulum is an angular harmonic oscillation. Moment of Inertia of a body is the reluctance to change its state of rest or uniform circular motion. A body whose moment of inertia I about an axis is known, is made to oscillate about the same axis, corresponding period T is noted. The ratio $\frac{I}{T^2}$ is a constant for different bodies and different axes **as long as the dimension of the suspension wire remains the same.** For a torsion pendulum $T=2\pi\sqrt{\frac{I}{C}}$ where C is the couple per unit twist of the wire and it is a constant. Hence, $I/T^2=C/4\pi^2$ is a constant and $C= 8\pi nr^4/2L$

Formula:

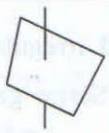
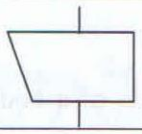
(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I'

$I_\alpha=\left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2=$ 3.064 x 10^-3 Kgm^2

(b) Moment of inertia of the irregular body about the axis through the CG and parallel to its plane

$I_\beta=\left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2=$ 2.157 x 10^-3 Kgm^2

(c) Rigidity modulus $n=\left(\frac{8\pi L}{r^4}\right)\left(\frac{I}{T^2}\right)$ where r is the radius and L is the length of the wire.

Axis through CG	Time for 10 sec. (s)	Mean time(t) for 10 oscillations in s	Period T=t/10 s
	1. 39.31 2. 39.56 3. 39.46	$\frac{118.33}{3} = 39.44$	$T_\alpha = 3.944$
	1. 33.01 2. 33.22 3. 33.06	$\frac{99.29}{3} = 33.09$	$T_\beta = 3.309$

Determination of Moment of Inertia of Irregular body

Formulae:

(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I

$$I_\alpha = \left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2 \quad Kgm^2$$

(b) Moment of inertia of the irregular body about the axis through the CG and parallel to its plane

$$I_\beta = \left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2 \quad Kgm^2$$

Determination of rigidity modulus of the material of the wire

Radius of the wire (r) using screw gauge.

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{1}{1} = 1 \text{ mm}$$

$$\text{L.C} = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) = -9.

Paste the data sheets here

Procedure:

- Measure the dimensions of the given circular and rectangular discs.
- Clamp one end of the wire through the chuck nut to a regular disc and other end to the top end of the retard stand.
- Twist the wire through a small angle and then let free so that the body executes torsional oscillations **(The oscillations should be in a horizontal plane. Arrest the side ward movement or wobbling if any)**.
- For each configuration of the pendulum, note down the time taken for 10 oscillations and repeat the process thrice. Tabulate this in table 1.
- Calculate mean time (t) for 10 oscillations and hence the time period T then find $\frac{I}{T^2}$ for each axis.
- Follow the same procedure for two different axes of the irregular body, determine the average period of oscillation for two different axes and tabulate values in table 2.
- Find out the moment of inertia of irregular body using given formulae.
- Measure the diameter of the wire using screw gauge and enter the readings in the tabular column.
- Calculate the average diameter and radius of the wire.
- Measure the length L of the wire between the check nuts.
- Calculate the rigidity modulus of the material of the wire using the given formula.

Paste the data sheets here

Trial No.	PSR(mm)	HSD	TR= PSR + (HSD- ZE)LC (mm)
1	0	65	$0+[65-(-9)]\times 0.01 = 0.74$
2	0	66	$0+[66-(-9)]\times 0.01 = 0.75$
3	0	64	$0+[64-(-9)]\times 0.01 = 0.73$
4	0	66	$0+[66-(-9)]\times 0.01 = 0.75$

Mean diameter of the wire, d = 0.742 mm 7.42×10^{-4} m

Mean Radius of the wire, r = 3.71×10^{-4} m

Length of the wire, L = 33.8×10^{-2} m

Rigidity modulus $n=\left(\frac{8\pi L}{r^4}\right)\left(\frac{I}{T^2}\right)$

$$= \frac{8\times3.14\times33.8\times10^{-2}}{(3.71\times10^{-4})^4}\times 1.97\times10^{-4}$$

$$= 8.333\times10^{10}\ N/M^2$$

Result:

1. The moment of inertia of irregular body about an axis perpendicular to the plane I_α = 3.064×10^{-3} Kgm^2.
2. The moment of inertia of irregular body about an axis parallel to the plane I_β = 2.167×10^{-3} Kgm^2.
3. Rigidity modulus of the material of the wire 'n' = 8.333×10^{10} N/m^2

CALCULATIONS:

$I_1 = \frac{MR^2}{2} = \frac{0.88(0.076)^2}{2} = 2.54\times10^{-3}\ kgm^2$

$\frac{I_1}{T_1^2} = \frac{2.54\times10^{-3}}{(3.568)^2} = 1.99\times10^{-4}\ kgm^2/s^2$

$I_2 = \frac{MR^2}{4} = \frac{0.88(0.076)^2}{4} = 1.27\times10^{-3}\ kgm^2$

$\frac{I_2}{T_2^2} = \frac{1.27\times10^{-3}}{(2.538)^2} = 1.97\times10^{-4}\ kgm^2/s^2$

$I_3 = \frac{M(L^2+B^2)}{12} = \frac{1.015[(0.15)^2+(0.1)^2]}{12} = 2.74\times10^{-3}\ kgm^2$

$\frac{I_3}{T_3^2} = \frac{2.74\times10^{-3}}{(3.763)^2} = 1.94\times10^{-4}\ kgm^2/s^2$

$I_4 = \frac{ML^2}{12} = \frac{1.015(0.15)^2}{12} = 1.90\times10^{-3}\ kgm^2$

$\frac{I_4}{T_4^2} = \frac{1.90\times10^{-3}}{(3.159)^2} = 1.97\times10^{-4}\ kgm^2/s^2$

$I_5 = \frac{MB^2}{12} = \frac{1.015(0.1)^2}{12} = 8.45\times10^{-4}\ kgm^2$

$\frac{I_5}{T_5^2} = \frac{8.45\times10^{-4}}{(2.065)^2} = 1.98\times10^{-4}\ kgm^2/s^2$

$\left(\frac{I}{T^2}\right)_{mean} = \frac{1.99\times10^{-4}+1.97\times10^{-4}+1.94\times10^{-4}+1.97\times10^{-4}+1.98\times10^{-4}}{5} = \frac{9.85\times10^{-4}}{5} = 1.97\times10^{-4}\ kgm^2/s^2$

$I_\alpha = \left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2 = 1.97\times10^{-4}\times(3.944)^2 = 3.064\times10^{-3}\ kgm^2$

$I_\beta = \left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2 = 1.97\times10^{-4}\times(3.309)^2 = 2.157\times10^{-3}\ kgm^2$

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Result:

1. The moment of inertia of irregular body about an axis perpendicular to the plane $I_\alpha = 3.064\times10^{-3}$ Kgm^2.

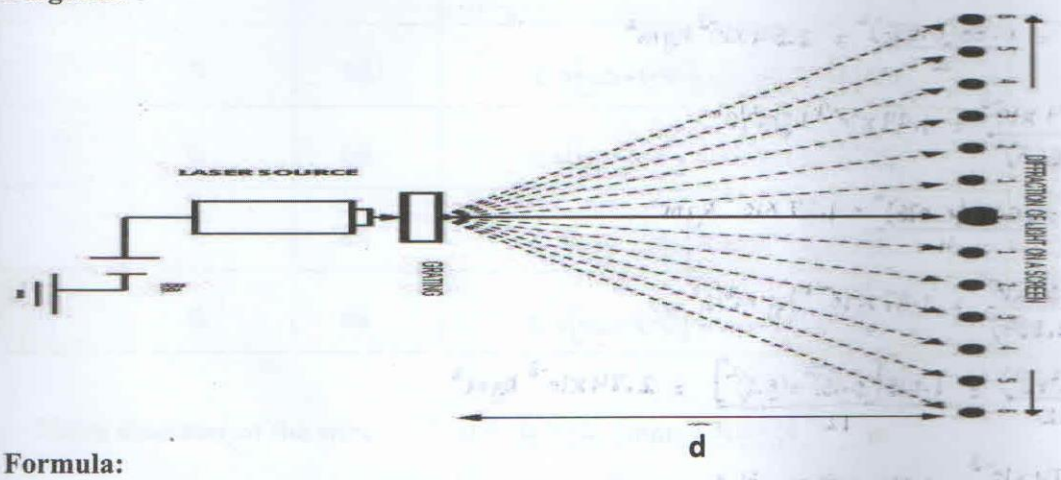
2. The moment of inertia of irregular body about an axis parallel to the plane $I_\beta = 2.157\times10^{-3}$ Kgm^2.

1. Rigidity modulus of the material of the wire 'n' = 8.333×10^{10} N/m^2

LASER DIFFRACTION

OBSERVATIONS:

Diagram:



Formula:

Wavelength of Laser source $\lambda = \dfrac{C \sin\theta_n}{n}$ ..... m

Where C is the grating constant, n is the order of the maximum, θ is the angle of diffraction

Grating Constant: $C = \dfrac{1\,\text{inch}}{\text{No. of lines (N) per inch}} = \dfrac{2.54\times10^{-2} m}{500} = 5.08\times10^{-5}\text{m}$

Distance between the grating and the screen, d = 1.185 m = 118.5 cm

Table:

Diffraction order (n)	Distance $2X_n$(cm)	Distance X_n(cm)	Diffraction angle (θ_n) $\theta_n = \tan^{-1}\left(\frac{X_n}{d}\right)$	Wavelength λ $\lambda = \frac{C\sin\theta_n}{n}$
1.	3.1	1.55	0.832	653.88 nm
2.	6.1	3.05	1.638	653.46 nm
3.	9.2	4.6	2.470	656.82 nm
4.	12.2	6.1	3.274	652.84 nm
5.	15.3	7.65	4.104	654.51 nm
6.	18.4	9.2	4.932	655.27 nm

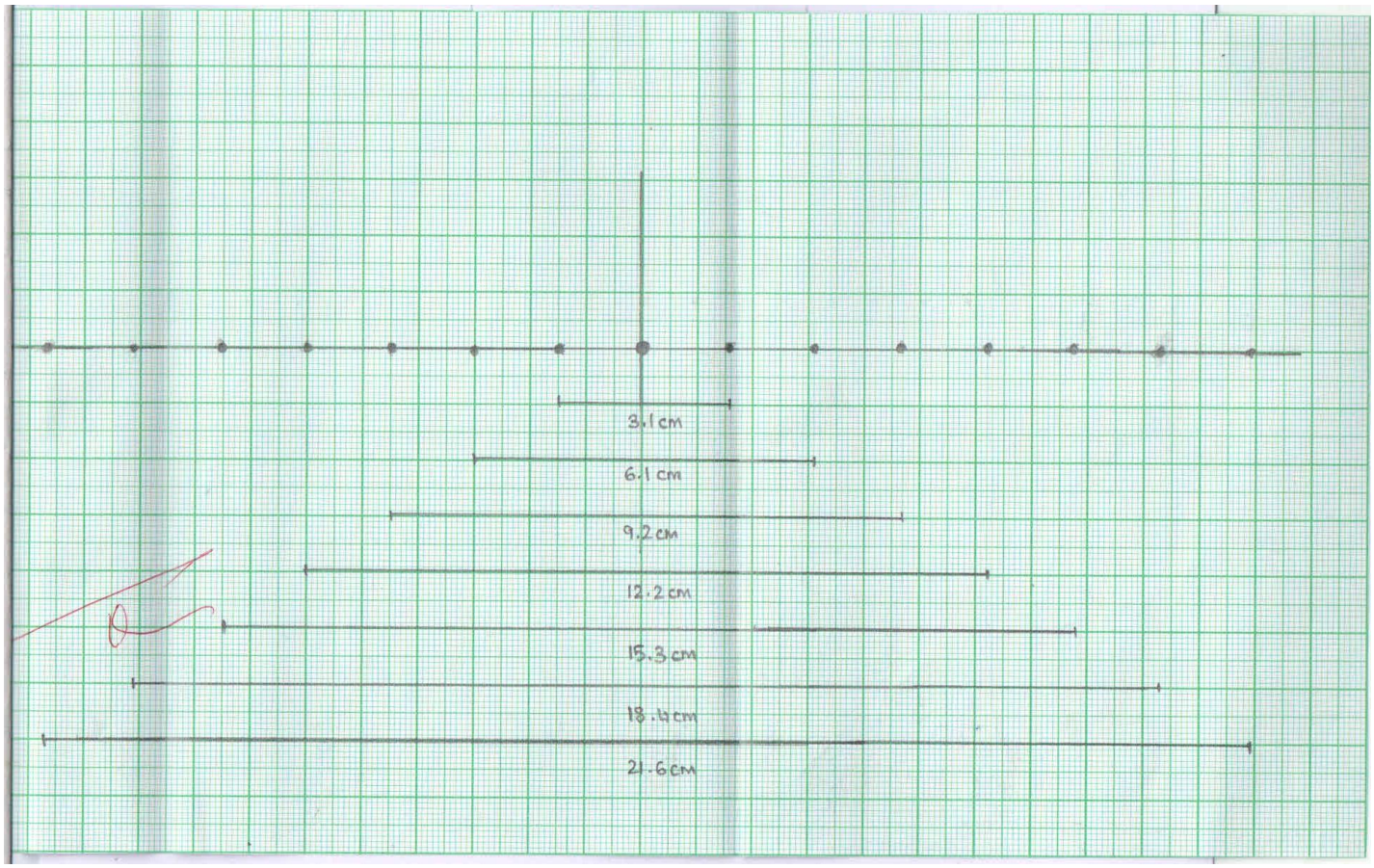
CALCULATIONS:

$\lambda_{avg} = \dfrac{653.88 + 653.46 + 656.82 + 652.84 + 654.51 + 655.27}{6}$

$= \dfrac{3926.76}{6}$

$= 654.46$ nm

Result: The wavelength of laser light is found to be 654.46 nm


3.1 cm
6.1 cm
9.2 cm
12.2 cm
15.3 cm
18.4 cm
21.6 cm

LASER DIFFRACTION

Experiment No: Date:

Aim: To determine the wavelength of a given laser beam

Apparatus and other required materials:
Laser source, Grating, Optical bench with accessories and metre scale etc.,

Principle: laser is a device, which gives a strong beam of coherent photons by stimulated emissions. The laser beam is highly monochromatic, coherent, directional and intense. The directionality of a laser beam is expressed in terms of full angle beam divergence. Divergence of a laser beam is defined as its spread with distance.

Formula:

Wavelength of Laser source, $\lambda = \frac{C \sin\theta_n}{n}$ ..... m

Where C is the grating constant: Distance between successive lines on the grating, n is the order of the maximum, θ is the angle of diffraction,

Procedure:

- Mount the laser on an upright and fix the upright at one end of the optical bench. Mount a screen on another upright and fix it at the other end of the optical bench.
- Mark four quadrants on a graph with 'O' as the origin and fix the graph sheet on the screen using pins. Place a laser source in front of the grating and adjust the position of the graph sheet, so that the centre of the laser spot coincides with the origin.
- Mount the grating on the grating stand such that the length of the grating is on the grating stand and move the stand closer to the laser source. Adjust the grating plane such that the diffraction pattern is along the horizontal on the screen with the central maximum is at the origin. Note down the distance'd' between grating and the screen.
- Mark the centres of the central maximum and secondary maxima on the graph sheet using pencil and remove the graph sheet from the stand. Measure the distance between the first order maxima on either side of the central maximum as $2X_1$, for the 2nd order maxima measure the distance as $2X_2$, continue this up to 4th order maxima.
- By using the grating constant C and the angle of diffraction θ_n, calculate the wavelength of laser light for all the orders. Finally find the average value of wavelength.

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Result: The wavelength of laser light is found to be.....654.46.....nm

VOLUME RESONATOR

OBSERVATIONS

Experimental setup:

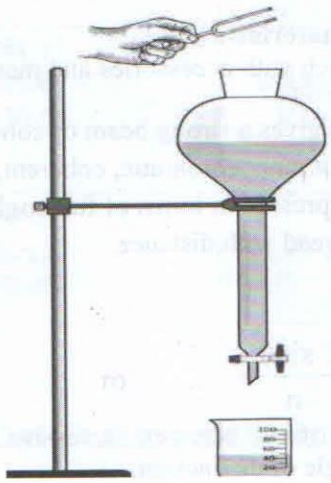


Table:

Serial Number	Frequency of the tuning fork f in Hz	Volume of resonating air 'V' in m^3			f^2V in m^3s^{-2}	$(f^2V)_{avg}$
		Trial 1	Trial 2	Mean		
1.	512	80×10^{-6}	80×10^{-6}	80×10^{-6}	20.97	18.036
2.	288	190×10^{-6}	195×10^{-6}	192.5×10^{-6}	16.79	18.036
3.	256	235×10^{-6}	240×10^{-6}	237.5×10^{-6}	16.35	18.036
4.	Unknown	110×10^{-6}	105×10^{-6}	$V_x = 107.5\times10^{-6}$		

Formulae: The unknown frequency of a tuning fork is given by,

$$f_x = \sqrt{\frac{(f^2V)_{avg}}{V_x}} = 409.61 \ldots Hz$$

Where, f_x = unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_x = volume of air resonating with the fork of unknown frequency.

Calculations:

$(f^2V)_1 = (512)^2 \times 80\times10^{-6} = 20.97\ m^3/s$

$(f^2V)_2 = (288)^2 \times 192.5\times10^{-6} = 16.79\ m^3/s$

$(f^2V)_3 = (256)^2 \times 237.5\times10^{-6} = 16.35\ m^3/s$

$(f^2V)_{avg} = \frac{(f^2V)_1 + (f^2V)_2 + (f^2V)_3}{3} = \frac{20.97 + 16.79 + 16.35}{3} = \frac{54.11}{3} = 18.036\ m^3/s$

$f_x = \sqrt{\frac{(f^2V)_{avg}}{V_x}} = \sqrt{\frac{18.036}{107.5\times10^{-6}}} = \sqrt{0.1677825\times10^6} = 409.61\ Hz$

VOLUME RESONATOR

Experiment No: Date:

Aim: To determine the frequency of a tuning fork using volume resonator.

Apparatus: Aspirator bottle, tuning forks, rubber pad, beaker, measuring jar.

Principle: A resonator is a device used to analyse complex note of sound i.e. to determine particular frequencies present in the given note. The resonator is a large cylindrical vessel made of glass with a narrow neck through which a stream of air is set vibrating by running in and out of it. When a tuning fork of particular frequency is excited and held near the mouth of the resonator. The resonator exhibits a sharp resonance, when cavity of air with an opening which resonates at a natural frequency when the air is excited. The relationship of the resonate frequency to the volume of the cavity is given by the following equation.

$$f_x = \sqrt{\frac{(f^2V)_{avg}}{V_x}} Hz$$

Where, f_x = unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_x = volume of air resonating with the fork of unknown frequency.

Procedure: The aspirator bottle is filled with water. The tuning fork of known frequency is excited and it is held above the mouth of the aspirator bottle as shown in the figure. At the same time, water in the bottle is allowed to run out of aspirator bottle by opening the pinch cock attached to the out let tube. Water is allowed to flow out till the air in the resonator is in resonance with the tuning fork used. In the correct position of resonance, a note of maximum intensity will be heard. At the same moment the flow of water should be stopped using the pinch cock. The volume 'V' of water is collected in a beaker and measured using measuring jar. The volume is equal to the volume of air in the resonator, which is in resonance with the tuning fork, when vibrated. The experiment is repeated for the same tuning fork and the mean value of volume 'V' is found and hence f^2V is calculated.

The experiment is repeated for different tuning fork as above and average f^2V is calculated. The volume of air V_x is found out with the fork of unknown frequency f_x and unknown frequency of the tuning fork is calculated using the above formula.

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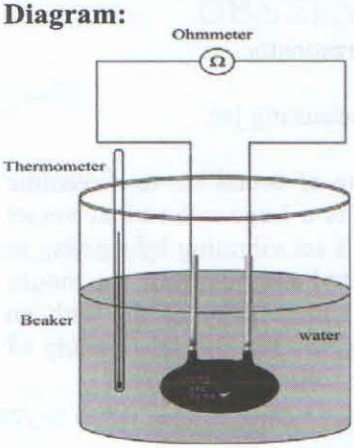
Result: The unknown frequency of given tuning fork = 409.61 Hz.

Note: Start the experiment by choosing higher frequency tuning fork and proceed the experiment in decreasing order of their frequencies.

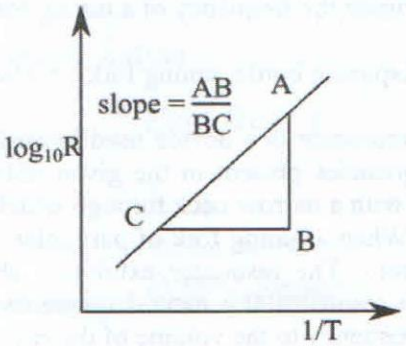
BAND GAP OF A THERMISTOR

OBSERVATIONS:

Diagram:



Model Graph:



Formula: $E_g = \dfrac{4.606 \times k \times m}{1.6 \times 10^{-19}}$ eV

Where E_g = Energy gap of a given thermistor in eV

k = Boltzmann constant = 1.381×10^{-23} J/K

m = Slope of the graph

Table:

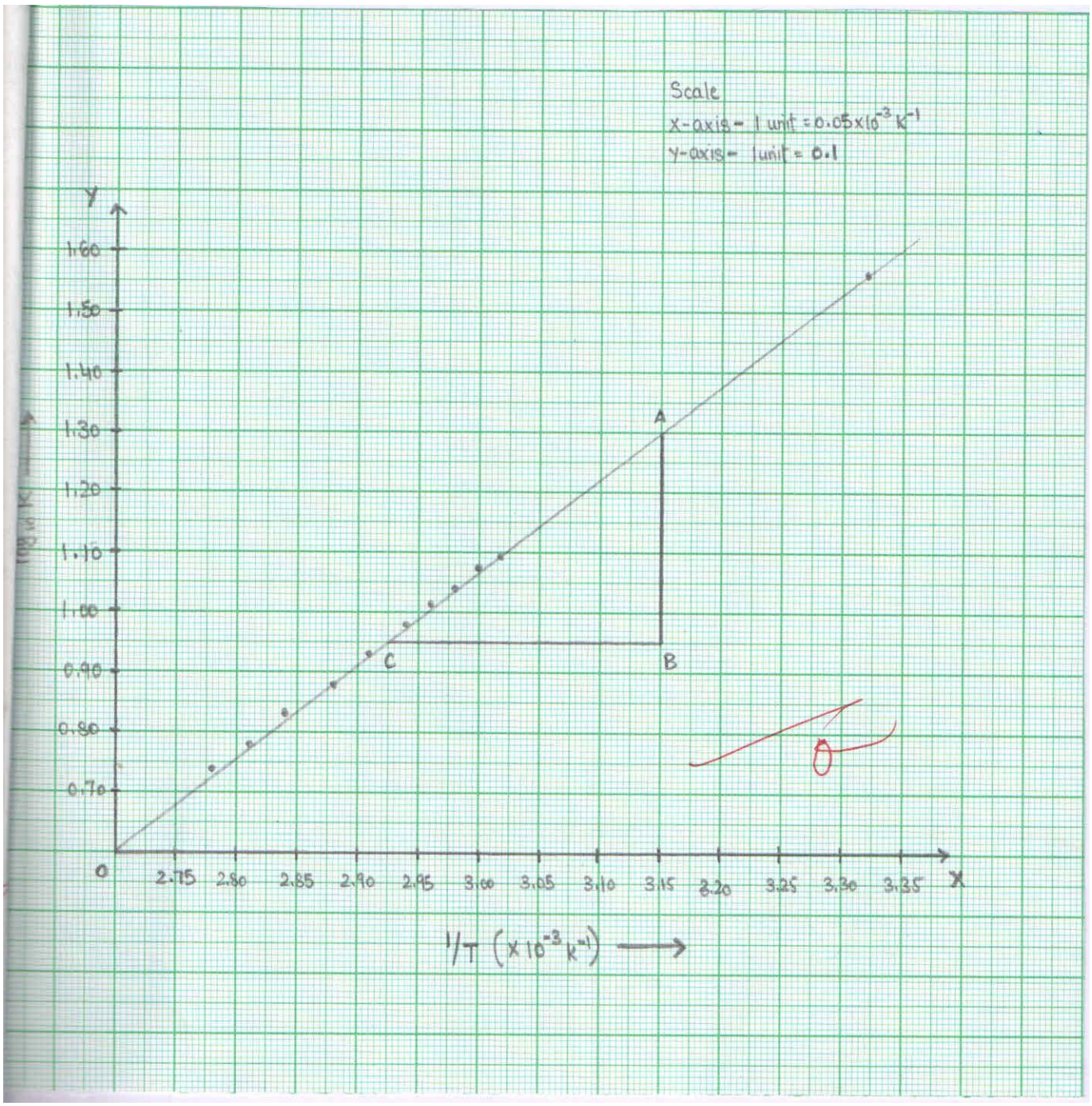
Sl. No	Temp t °C	Temp T(K)	R (Ω)	log R	1/T ($\times 10^{-3}$ K^{-1})
Room Temp	27.9	300.9	36.7	1.564	3.32×10^{-3}
1.	86	359	5.5	0.740	2.78×10^{-3}
2.	82	355	6.1	0.785	2.81×10^{-3}
3.	78	351	6.8	0.882	2.84×10^{-3}
4.	74	347	7.6	0.880	2.88×10^{-3}
5.	70	343	8.6	0.934	2.91×10^{-3}
6.	66	339	9.7	0.986	2.94×10^{-3}
7.	64	337	10.2	1.008	2.96×10^{-3}
8.	62	335	10.9	1.037	2.98×10^{-3}
9.	60	333	11.5	1.060	3.00×10^{-3}
10.	58	331	12.3	1.089	3.02×10^{-3}

CALCULATIONS:

$$\text{Slope, } m = \frac{AB}{BC} = \frac{1.30 - 0.95}{(3.15 - 2.925) \times 10^{-3}} = \frac{0.35}{0.225 \times 10^{-3}} = 1.55 \times 10^{3}$$

$$E_g = \frac{4.606 \times K \times m}{1.6 \times 10^{-19}} = \frac{4.606 \times 1.381 \times 10^{-23} \times 1.55 \times 10^{3}}{1.6 \times 10^{-19}} = \frac{9.894 \times 10^{-20}}{1.6 \times 10^{-19}} = 0.618 \text{ eV}$$

Result: The energy gap (band gap) of the given thermistor is 0.618 eV.


Scale
x-axis - 1 unit = 0.05 x 10^-3 K^-1
y-axis - 1 unit = 0.1
Y
1.60
1.50
1.40
1.30
1.20
1.10
1.00
0.90
0.80
0.70
A
B
C
O
2.75
2.80
2.85
2.90
2.95
3.00
3.05
3.10
3.15
3.20
3.25
3.30
3.35
X
1/T (x 10^-3 K^-1) ⟶

BAND GAP OF A THERMISTOR

Experiment No: Date:

Aim: To determine the energy gap (E_g) of a Thermistor.

Apparatus: Glass beaker, Thermistor, Multi meter, Thermometer.

Principle: A thermistor is a thermally sensitive resistor. Thermistors are made of semiconducting materials such as oxides of Nickel, Cobalt, Manganese and Zinc. They are available in the form of beads, rods and discs.

The variation of resistance of thermistor is given by $R = ae^{\frac{b}{T}}$ where 'a' and 'b' are constants for a given thermistor, b is a measure of the band gap. The resistance of thermistor decreases exponentially with rise in temperature. At absolute zero all the electrons in the thermistor are in valence band and conduction band is empty. As the temperature increases electrons jump to conduction band and the conductivity increases and hence resistance decreases. By measuring the resistance of thermistor at different temperatures the energy gap is determined.

Formula: $E_g = \frac{4.606 \times k \times m}{1.6 \times 10^{-19}} eV$

Where, E_g = Energy gap of a given thermistor in eV, k = Boltzmann constant = 1.381 x 10^{-23} J/K .

m = Slope of the graph of log R vs 1/T.

Procedure:

- Make the circuit connection as shown in the figure.
- Keep the multi meter in resistance mode (200 Ω range).
- Insert the thermometer in a beaker containing tap water, thermistor and note down the resistance at room temperature.
- Immerse the thermistor in hot water at about 90°C.
- Note down the resistance of the thermistor for every decrement of 1°C in the beginning and a decrement of 2^0 up to 60°C.
- Plot the graph of log R versus 1/T and calculate the slope 'm'.
- Calculate the energy gap of a given thermistor using relevant formula.

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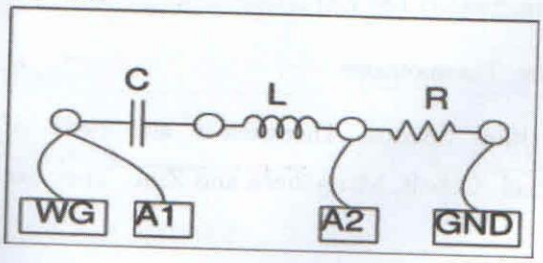
Result: The energy gap (band gap) of the given thermistor is 0.618 eV.

23/11

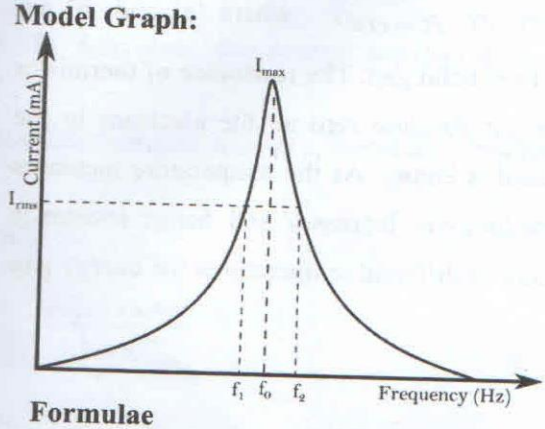
SERIES L-C-R CIRCUIT

OBSERVATIONS:

Circuit:



Model Graph:



Graph plotting with GNUPLOT

Go to respective folder in which the file is saved. Right click at the empty area, select **'open in terminal'** and Run the following commands:

```
set timestamp
set title 'Series LCR circuit-Raj'
set xlabel 'Frequency in Hz'
set ylabel 'Current in mA'
plot 'lcr.txt' w lp
```

Marking points on the graph

- Take the cursor to the tip of the curve
- Press the scroll button on the mouse
- Note down the resonant frequency and I_{max}
- Find out I_{RMS}
- Press the scroll button on the mouse at I_{RMS}
- (Both at left and right side of the curve)
- Note down f_1 and f_2
- Export the file as PDF

Formulae

I_{max} = 30.84 mA

$I_{rms} = I_{max}/\sqrt{2}$ = 21.80 mA

Resonant Frequency f_0 = 199.77 Hz

Lower Cut off frequency f_1 = 186.42 Hz

Upper Cut off frequency f_2 = 217.56 Hz

Band width $\Delta f = f_2 - f_1$ = 31.14 Hz

Quality Factor $Q = f_0/\Delta f$ = 6.122

Capacitance C = 0.474 μF

$$L = \frac{1}{4\pi^2 f_0^2 C} = 1.325 \text{ H}$$

Calculations:

$I_{rms} = \frac{I_{max}}{\sqrt{2}} = \frac{30.84}{\sqrt{2}} = 21.80$ mA

$\Delta f = f_2 - f_1 = 217.56 - 186.42 = 31.14$ Hz

$Q = \frac{f_0}{\Delta f} = \frac{199.77}{31.14} = 6.122$

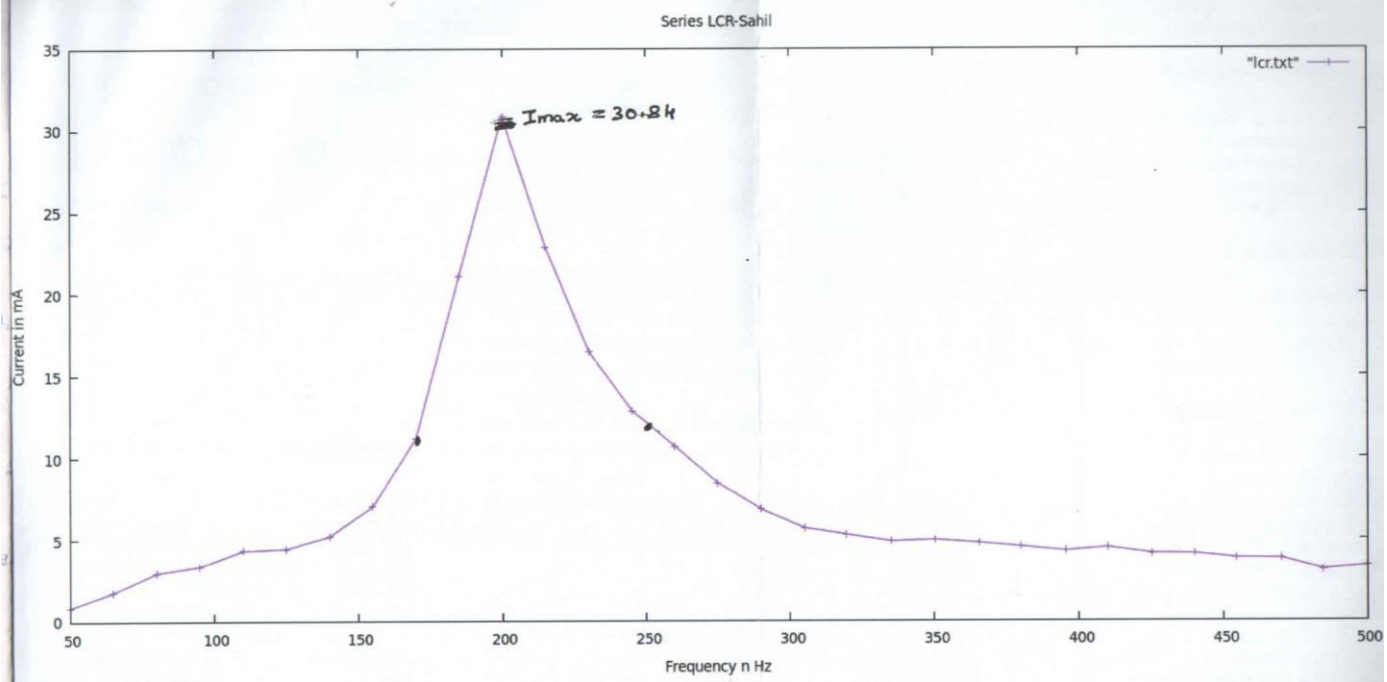
$L = \frac{1}{4\pi^2 f_0^2 C} = \frac{1}{4\times(3.14)^2 \times (199.77)^2 \times 0.474\times10^{-6}}$

$= 1.325$ H

Result:

1.	Resonant frequency of the circuit f_0	199.77	Hz
2.	Self-inductance of the given coil L	1.325	H
3.	Quality factor from graph Q	6.122	
4.	Band width $\Delta f = f_2 - f_1$	31.14	Hz

SERIES L-C-R CIRCUIT



Wed Dec 13 16:40:33 2023

SERIES L-C-R CIRCUIT

Experiment No: Date:

Aim: To study the frequency response of LCR circuit and determination of a) Self-inductance of the given coil, b) Quality factor (Q- value) and c) Band-width

Apparatus: expEYES-17 kit, Resistance box, Capacitor, Inductor

Principle: In a series LCR circuit, as the frequency increases inductive reactance X_L of the circuit increases and capacitive reactance X_c decreases. At resonant frequency f_0. The inductive reactance and capacitive reactance are equal hence the total impedance of the circuit is minimum and there by the current is maximum. Therefore, at resonance $X_l = X_C$,
$2\pi f_0 L = 1/2\pi f_0 C \rightarrow L = 1/4\pi^2 f_0^2 C$

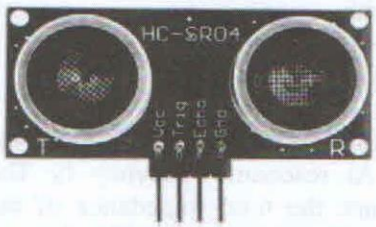
Where f_0 is the resonant frequency, L is the inductance of the coil and C is the value of capacitance.
The property of a reactive circuit to store energy is expressed in terms of quality factor or 'Q' factor. It is a figure of merit that enables us to compare different coils. It is defined as follows
$Q = \omega_0$ x energy stored / average power dissipated. $[Q=f_0/f_2-f_1]$
The band width Δf is the difference in frequencies corresponding to current $I_{max}/\sqrt{2}$

Procedure:

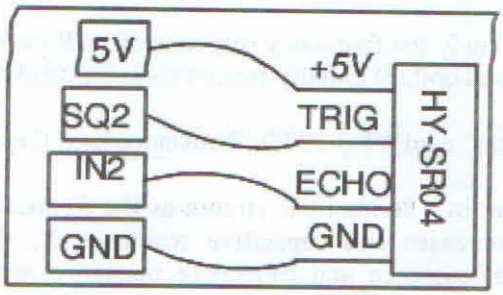
1. Make the connections as per the circuit diagram.
2. Select the frequency range in the right side of GUI window.
3. Enter the value of resistance in the box (keys unplugged in the resistance box).
4. Enter the frequency interval steps in the GUI window.
5. Click on START button to plot the current versus frequency.
6. Save the data by clicking on the Sava Data button.
7. A graph of frequency versus current is plotted using GNUPLOT software, this graph is called frequency response of the L C R circuit.
8. The frequency corresponding to maximum current I_{max} is called resonant frequency and is denoted by f_0. Corresponding to a current $I = I_{max}/\sqrt{2}$ from the current axis, a horizontal line is drawn such that it cuts the graph at points A and B. From the points A and B vertical lines are drawn to cut the frequency axis at f_1 and f_2 respectively. These frequencies are lower cut off and upper cut off frequencies (or half power frequencies). The frequencies f_0, f_1 and f_2 are noted. Self-inductance of the given coil, Quality factor and Band width Δf are calculated using the relevant formulae.
9. Note down the data points from the txt file.
10. Plot the graph on the graph sheet using the data points and find out f_1, f_0 and f_2.

SPRING CONSTANT

OBSERVATIONS

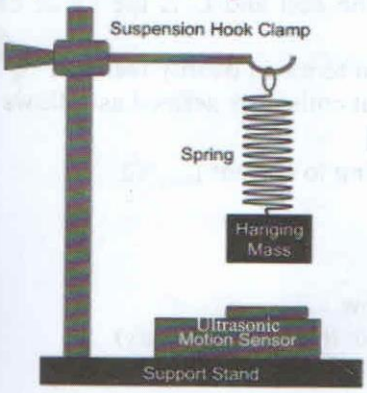


Ultrasonic sensor (HC-SR04)

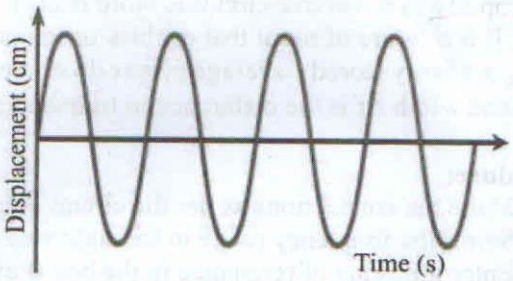


Connections with Exp-eyes device

Experimental setup



Model graph



Formula: Spring constant is given by $K = 4\pi^2 \frac{m}{T^2}$ N/m

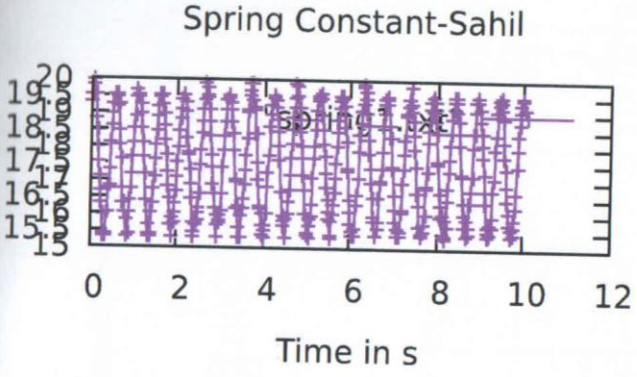
Where, *m* is the mass of the load (kg). T is the time period (s) and K is the spring constant N/m.

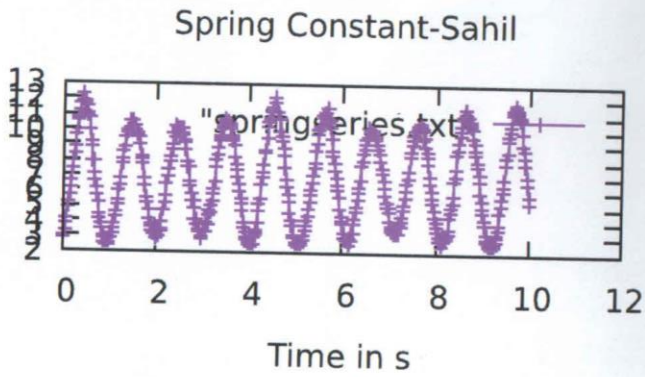
Graph plotting with GNUPLOT

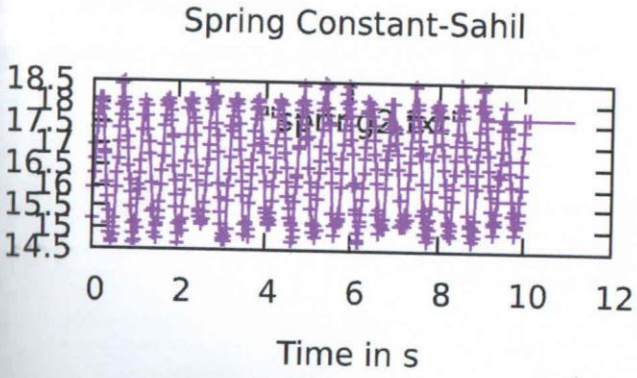
Go to respective folder in which the file is saved.

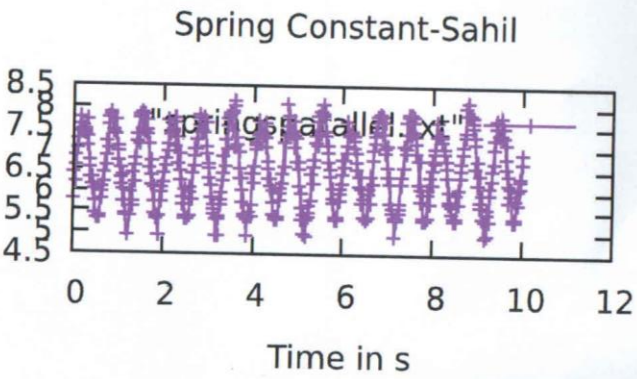
Right click at the empty area, select **'open in terminal'** and Run the following commands:

```
set timestamp
set title 'Spring Constant-Raj'
set xlabel 'Time in S'
set ylabel 'Displacement in cm'
set multiplot layout 2,2 columns
plot 'spring1.txt' w lp
plot 'spring2.txt' w lp
plot 'springseries.txt' w lp
plot 'springparallel.txt' w lp
```









Nov 24 16:46:08 2023

SPRING CONSTANT USING ULTRASONIC SENSOR

Experiment No: Date:

Aim: Determine spring constant for the given spring using ultrasonic sensor.

Apparatus: springs, weight hanger, slotted weights, stop watch.

Principle: Spring constant (or force constant) of a spring is given by $K = \frac{\text{Restoring Force}}{\text{Extension}}$ N/m

Spring constant is the restoring force per unit extension in the spring. Its value is determined by the elastic properties of the spring. Elastic materials are those which retain their original dimensions after the removal of deforming forces. When material is subjected to stress, strain is produced. The restoring force is always directed opposite to the displacement. When the mass is displaced through a small distance and then released, it undergoes simple harmonic motion.

The time period **T** of oscillations of a spring is given by the relation,

$$T = 2\pi \sqrt{\frac{m}{K}} \text{ s}$$

By finding the time period (T) spring constant K can be determined.

Procedure:

1. Suspend one end of the given spring to a rigid support and add the slotted weights at the other end. And note down the mass of added slotted weights and weight hanger (m_1).
2. Below the load, Place an ultrasonic sensor (HC-SR04) after making the connections with the expEYES-17 hardware.
3. Keep the distance between the ultrasonic sensor and the load as 6cm.
4. Interface the experimental circuit with the CPU and click on mechanics tab and select distance by HC-SR04 echo module experiment.
5. On the screen enter the y axis value as 10cm and time as 10 seconds.
6. Pull the load slightly downwards and then release it gently so that it is set into oscillations in a vertical plane about its mean position.
7. Click on the start button. The graph will be generated on the screen.
8. Click on the 'fit curve using sine' button to generate a sine curve on the existing graph.
9. Note down the frequency in Hz and find time period of the oscillation. And then compute the spring constant value (K_1) for the load (m_1).
10. Repeat the same procedure for another identical spring (K_2) for the load (m_2)
11. Connect two springs in series combination and repeat the above activity and find K_S value. Calculate K_{eff} for series and compute the error.
12. Connect two springs in parallel combination and repeat the above activity and find K_P value. Calculate K_{eff} for parallel and compute the error



SPRING CONSTANT

OBSERVATIONS

Table 1: Spring constant K_1 for the given material

(Mass of the hanger + Mass of the slotted weights) = m_1 = 0.05 Kg

Trial No	Frequency(Hz)	Time for one osc T_1 (s)	T_1^2	Spring constant $K_1=4\pi^2\frac{m_1}{T_1^2}$(N/m)
1	1.91	0.52	0.2704	7.29

Table2: Spring constant K_2 for the given material

(Mass of the hanger + Mass of the slotted weights) = m_2 = 0.05 Kg

Trial No	Frequency(Hz)	Time for one osc T_2 (s)	T_2^2	Spring constant $K_2=4\pi^2\frac{m_2}{T_2^2}$(N/m)
1	1.92	0.52	0.2704	7.29

Table 3: Spring constant K_S in series combination for the given material

(Mass of the hanger + Mass of the slotted weights) = m_s = 0.1 Kg

Trial No	Frequency(Hz)	Time for one osc T_S (s)	T_S^2	Spring constant $K_S=4\pi^2\frac{m_s}{T_s^2}$ (N/m)	$K_{eff}=\frac{K_1K_2}{K_1+K_2}$ (N/m)	Error = $K_{eff} \sim K_S$ (N/m)
1	0.97	1.03	1.0609	3.717	3.645	0.072

Table 4: Spring constant K_P in parallel combination for the given material

(Mass of the Scale + Mass of the hanger + Mass of the slotted weights) = m_P= 0.1672 Kg

Trial No	Frequency(Hz)	Time for one osc T_P (s)	T_P^2	Spring constant $K_P=4\pi^2\frac{m_p}{T_P^2}$ (N/m)	$K_p=K_1+K_2$ (N/m)	Error= $K_{eff} \sim K_P$ (N/m)
1	1.51	0.66	0.4356	14.674	14.58	0.094

Results:

a)	The spring constant for the given springs are	K_1= 7.29 N/m K_2= 7.29 N/m
b)	The spring constant in series combination is	Ks= 3.717 N/m
c)	The spring constant in parallel combination is	Kp= 14.674 N/m

About hardware and software:

The circuit consists of a microcontroller system, an ultrasonic sensor, Expeyes hardware and a spring mass system. A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. An ultrasonic sensor measures the distance of an object using sound waves. The ultrasonic sensor consists of 4 pins (Voltage pin, Trigger Pin, ECHO pin, GND pin).Connections are made as shown in the figure, voltage pin is connected to +5V of the expEYES-17; the trig pin of the sensor is connected to the SQ2 of the expEYES-17; the echo pin is connected to the IN2; GND pin is connected to the GND of the expeyes-17. The expeyes-17 interfaced with CPU of the computer. It combines the real-time measurement capability of micro-controllers with the ease and flexibility of Python programming language for data analysis and visualization.

CALCULATIONS:

$K_1 = \frac{4\pi^2 m_1}{T_1^2} = \frac{4\times(3.14)^2\times 0.05}{(0.2704)^2} = 7.29$ N/M

$K_2 = \frac{4\pi^2 m_2}{T_2^2} = \frac{4\times(3.14)^2\times 0.05}{(0.2704)^2} = 7.29$ N/M

$K_S = \frac{4\pi^2 m_S}{T_S^2} = \frac{4\times(3.14)^2\times 0.1}{(1.0609)^2} = 3.717$ N/M

$K_{eff} = \frac{K_1 K_2}{K_1 + K_2} = \frac{7.29\times 7.29}{7.29+7.29} = 3.645$ N/M

Error $= K_S - K_{eff} = 3.717 - 3.645 = 0.072$ N/M

$K_p = \frac{4\pi^2 m_p}{T_p^2} = \frac{4\times(3.14)^2\times(0.1672)}{(0.4356)} = 14.674$ N/M

$K_{eff} = K_1 + K_2 = 7.29 + 7.29 = 14.58$ N/M

Error $= K_p - K_{eff} = 14.674 - 14.58 = 0.094$ N/M

Good! Very neat!

30/30

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